

freshly updated report. Henry was a graduate student in Operations Research when John hired him. His specialty was a field called Computational Finance, which, as its name indicates, seems to focus solely on running computer programs overnight. Henry's income went from \$50,000 to \$600,000 in three years.

Most of the profit John generated for the institution was not attributable to the interest rate differential between the instruments described above. It came from the changes in the value of the securities John held, mostly because many other traders were acquiring them to imitate John's trading strategy (thus causing the price of these assets to rise). The interest rate differential was getting closer to what John believed was "fair value". John believed that the methods he used to calculate "fair value" were sound. He was backed by an entire department that helped him analyze and determine which bonds were attractive and offered capital appreciation potential. It was normal for him to be earning these large profits over time.

John made steady money for his employers, perhaps even better than steady. Every year the revenues he generated almost doubled as compared to the previous year. During his last year, his income experienced a quantum leap as he saw the capital allocated to his trades swell beyond his wildest expectations. His bonus check was for \$10 million (pretax, which would generate close to a \$5 million total tax bill). John's personal net worth reached \$1 million at the age of 32. By the age of 35 it had exceeded \$16 million. Most of it came from the accumulation of bonuses – but a sizeable share came from profits on his personal portfolio. Of the \$16 million, about \$14 million he insisted in keeping invested in his business. They allowed him, thanks to the leverage (i.e., use of borrowed money), to keep a portfolio of \$50 million involved in his trades, with \$36 million borrowed from the bank. The effect of the leverage is that a small loss would be compounded and would wipe him out.

It took only a few days for the \$14 million to turn into thin air – and for John to lose his job at the same time. It all happened during the summer of 1998, with the meltdown of high-yield bond values. Markets went into a volatile phase in which nearly everything he had invested in went against him *at the same time*. His hedges no longer worked out. He was mad at Henry for not having figured out that these events could happen. Perhaps there was a bug in the program.

His reaction to the first losses was, characteristically, to ignore the market. "One would go crazy if one were to listen to the mood swings of the market", he said. What he meant by that statement was that the "noise" was mean reverting, and would likely be offset by "noise" in the opposite direction. That was the translation in plain English of what Henry explained to him. But the "noise" kept adding up in the same direction.

As in a biblical cycle, it took seven years to make John a hero and just seven days to make him an idiot. John is now a pariah; he is out of a job and his telephone calls are not returned. Many of his friends were in the same situation. How? With all that information available to him, his perfect track record (and therefore, in his eyes, an above average intelligence and skill-set) and the benefit of sophisticated mathematics, how could he have failed? Is it perhaps possible that he forgot about the shadowy figure of randomness?

It took a long time for John to figure out what had happened, owing to the rapidity with which the events unfolded and his state of shellshock. The dip in the market was not very large. It was just that his leverage was enormous. What was more shocking for him was that all their calculations gave the event a probability of 1 in 1,000,000,000,000,000,000,000 years. Henry called that a "ten sigma" event. The fact that Henry doubled the odds did not seem to matter. It made the probability 2 in 1,000,000,000,000,000,000,000 years.

When will John recover from the ordeal? Probably never. The reason is not because John lost money. Losing money is something good traders are accustomed to. It is because he blew up; he lost more than he planned to lose. His personal confidence was wiped out. But there is another reason why John may never recover. The reason is that John was never a trader in the first place. He is one of those people who happened to be there when it all happened.

Following the incident, John regarded himself "ruined"; yet his net worth is still close to \$1 million, which could be the envy of more than 99.9% of the inhabitants of our planet. Yet there is a difference between a wealth level reached from *above* and a wealth reached from *below*. The road from \$16 million to \$1 million is not as pleasant as the one from 0 to \$1 million. In addition, John is full of shame; he still worries about running into old friends on the street.